FABIAN ALVAREZ-PRIMO

OPERATIONS LEADER | REGULATED ENVIRONMENTS | BIOMANUFACTURING

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Executive Summary

Operations leader with PhD in biomedical engineering and proven track record scaling regulated processes. Currently managing **60+ team** operations at Amazon with **99.7% quality compliance**. Deep expertise in CLIA/COLA-regulated environments, GMP-adjacent operations, and automation—bridging scientific rigor with operational excellence for biomanufacturing scale-up.

Core Competencies

Regulated Operations (CLIA, COLA, GMP-adjacent)	Team Leadership & Development (60+ Direct Reports)	Quality Systems & Compliance
Process Optimization & Lean Manufacturing	Automation & Data Analytics (Python, R, SQL)	Cross-functional Collaboration
Scientific & Technical Depth (PhD, 8 Publications)	Biomanufacturing Scale-up & cGMP Processes	KPI Development & Performance Monitoring

Professional Experience

Operations Manager II | Amazon | Savannah, GA | 2024 – Present

- Manage operations team of **60+ associates** across a compressed 4-day schedule in a high-volume fulfillment environment, maintaining **99.7% quality compliance**.
- Implemented Python-based labor forecasting models, reducing scheduling variance and optimizing resource allocation across shifts.
- Led cross-functional process improvement initiatives applying Lean methodologies, driving measurable efficiency gains and waste reduction.
- Established quality control protocols and real-time KPI dashboards for performance monitoring across all operational metrics.
- Developed and delivered team training programs improving onboarding efficiency and accelerating time-to-productivity; trained **15 Process Assistants**, managing up to **9 concurrently**, building a structured PA development pipeline where none previously existed.
- Engineered a three-layer Ship Dock intelligence system — a Tampermonkey browser automation script scraping Amazon's Foresight WIP tool across paginated chute/CPT data, a real-time Google Sheets analytics layer tracking recirculation root causes, staffing ratios, DPMO, and hourly throughput, and a v3.0 staffing and rate management tool — replacing entirely manual workflows with an integrated data pipeline purpose-built around gaps in existing tooling.
- Independently engineered a suite of Google Sheets and Apps Script production tools deployed across active fulfillment operations: Thrive PA review automation (narrative generation from performance data), FORESIGHT volume allocation formula (CART/FLATS split logic across paginated chute data), and RTB Comment Generator (pandas-based UPH analysis with rate-dilution logic for new-hire cohorts) — replacing entirely manual workflows with structured, repeatable data pipelines.
- Drove an **82% improvement in dock rate (220 → 400+ units/hr)** and reduced recirculation from **5% to below 2%** through data-driven staffing models, chute-level congestion diagnostics, and process standardization across material paths and Vendor Returns operations — the latter built from scratch with no prior operational structure at site.
- Automated associate scheduling and productivity tracking via Thrive, eliminating manual entry bottlenecks and enabling real-time visibility into labor utilization and shift performance across the dock.
- Identified systemic tooling gaps in ship dock operations, independently engineered cross-functional solutions (automation + analytics + staffing frameworks), and drove org-wide adoption — exemplifying the Invent & Simplify arc: gap identified → bridge built → org caught up.

Ventures & Technical Projects

LabOps.AI | Founder & Technical Lead | 2024 – Present

- AI-native literature intelligence platform for biomanufacturing facilities targeting cell and gene therapy operators.
- Architected end-to-end pipeline: bioRxiv/PubMed ingestion → Unpaywall PDF resolution → PyMuPDF parsing → ChromaDB vector storage → Gemini 2.5 Flash synthesis with structured JSON extraction of Critical Quality Attributes (CQAs) and Critical Process Parameters (CPPs).
- Tech stack: Python 3.13, Gemini 2.5 Flash, ChromaDB, SQLite (WAL mode), PyMuPDF, Unpaywall API.
- Pre-seed stage; literature intelligence layer production-ready.

Post-doctoral Research Fellow | Texas Tech University Health Sciences Center (TTUHSCEP) | El Paso, TX | 2022 – 2023

- Led iPSC reprogramming project, directing non-integrating conversion of human PBMCs into clinically relevant induced Pluripotent Stem Cells under **cGMP-compliant** protocols for neuropsychiatric disease modeling.
- Oversaw biobanking operations including sample intake, processing, storage, and chain-of-custody documentation to maintain regulatory integrity.
- Managed multi-experiment research data pipelines, applying statistical analysis (R, Python) for transcriptomic and biomarker datasets.
- Collaborated across neuroscience, clinical, and molecular biology teams to coordinate study timelines, protocols, and data deliverables.

Clinical Laboratory Scientist & Supervisor | West Texas Pain Institute (WTPI) | El Paso, TX | 2021 – 2022

- Oversaw Clinical Laboratory Testing operations processing **1,300+ patient samples monthly**, leading a team of **9 laboratory professionals** within a Pain Medicine practice.
- Ensured full **CLIA and COLA regulatory compliance** across all laboratory operations, including DHHS guideline adherence and Drug of Abuse protocols.
- Operated and maintained advanced analytical instrumentation including LC/MS confirmatory systems, Enzyme Immunoassay platforms (CEDIA, DRI, QMS, EIA), and DNA next-generation sequencing.
- Improved laboratory workflow through rigorous documentation, OSHA protocol enforcement, and data processing software implementation.
- Managed LIS bioinformatics system (Stratus DX) and updated patient EMR records (eClinicalWorks), ensuring accuracy and compliance.

Doctoral Research Assistant | Inspired Materials & Stem Cell Tissue Engineering Lab (IMSTEL), UTEP | El Paso, TX | 2017 – 2020

- Mentored and trained **30+ undergraduate students** on electron microscopy and 3D bioprinting, developing structured onboarding curricula and hands-on training protocols.
- Led multi-year R&D project designing electrically conductive cardiovascular tissue composites for cell therapy applications, in collaboration with Rice University.
- Managed cross-functional lab operations—coordinating equipment schedules, reagent procurement, SOPs, and data management across **5+ concurrent projects**.
- Achieved **30% improvement** in 3D printer manufacturing precision through hardware modifications.

Education

Ph.D. in Biomedical Materials Science & Engineering | The University of Texas at El Paso | 2016 – 2020

B.S. in Materials Engineering | The University of Texas at El Paso | 2010 – 2015

Publications & Professional Honors

8 peer-reviewed publications in biomaterials, tissue engineering, and neuropsychiatry (available upon request).

Tau Beta Pi Engineering Honor Society | Alpha Chi National Honor Society | Alpha Sigma Mu Materials Honor Society | Society for Biomaterials (President, 2 terms)